

Ghazi Muhammad Abdullah

Domain: Artificial Intelligence & Machine Learning

Submitted to : Mr. Mohazin Zahid

Experiment 1

Lab Tasks

Experiment 1.py ×

```
1      # Take input from the user
2      num1 = float(input("Enter first number: "))
3      num2 = float(input("Enter second number: "))
4
5      # Perform calculations
6      print("Sum =", num1 + num2)
7      print("Difference =", num1 - num2)
8      print("Product =", num1 * num2)
9      print("Quotient =", num1 / num2)
```

```
11     # Take temperature in Celsius
12     celsius = float(input("Enter temperature in Celsius: "))
13
14     # Convert to Fahrenheit
15     fahrenheit = (celsius * 9 / 5) + 32
16
17     # Display result
18     print("Temperature in Fahrenheit =", fahrenheit)
```

```
20     # Take user's name
21     name = input("Enter your name: ")
22
23     # Print greeting five times
24     for i in range(5):
25         print("Hello", name)
```

Viva Questions

1. Artificial Intelligence (AI) is the broad field of creating machines that can perform tasks requiring human intelligence. Machine Learning (ML) is a subset of AI that enables computers to learn from data without explicit programming. Deep Learning (DL) is a subset of ML that uses neural networks with multiple layers to solve complex problems such as image and speech recognition.
2. Anaconda is used because it provides Python along with many pre-installed libraries used in data science and Machine Learning. It also includes tools like Jupyter Notebook and Anaconda Navigator, making development easier.
3. A Jupyter Notebook cell is a section where Python code or text can be written. A code cell is executed by pressing **Shift + Enter**.
4. The `input()` function returns the user's input as a string. We use `int()` or `float()` to convert the input into a numeric data type when performing calculations.
5.
 - **NumPy** is used for numerical computations and array operations.
 - **Pandas** is used for data manipulation and analysis.
 - **Scikit-learn** is used for building and training Machine Learning models.
6. Jupyter Notebook runs locally on a computer and requires Python to be installed, whereas Google Colab is a cloud-based platform that runs in a web browser and provides free access to GPUs and TPUs for Machine Learning tasks.

MCQ's

- Q1.** Machine Learning is a subset of:
- (b) Artificial Intelligence**
- Q2.** Which tool installs Python plus data-science libraries in one package?
- (b) Anaconda**
- Q3.** Which shortcut runs a Jupyter cell?
- (b) Shift+Enter**

Q4. The input() function returns a value of type:

(c) string

Q5. Which library is used for numerical arrays?

(b) NumPy

Intern Name: Ghazi Muhammad Abdullah

Supervisor Name: Mr. Mohazin Zahid

Date: 06 July, 2026

Domain: AI/ML Intern

Intern Signature:

A handwritten signature in black ink, appearing to read 'Ghazi', with a stylized flourish underneath.